

FACTORS REGULATING (AFFECTING) GFR

Headings

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1. Renal Blood Flow (RBF)

▼ 2. Tubuloglomerular Feedback (TGF)

1. When GFR / NaCl conc. inc. in filtrate

2. When GFR / NaCl conc. dec. in filtrate

Tubuloglomerular Feedback Sensitivity

3. Glomerular Capillary Hydrostatic Pressure (P_{GC})

4. Colloid Osmotic / Oncotic Pressure (π_{GC})

5. Hydrostatic Pressure in Bowman's Capsule (P_{BS})

6. Constriction of Afferent Arteriole

7. Constriction of Efferent Arteriole

8. Systemic Arterial Blood Pressure

9. Sympathetic Stimulation

10. Surface Area of Capillary Membrane

11. Permeability of Capillary Membrane

12. Contraction of Glomerular Mesangial Cells

13. Hormonal and Other Vasoactive Factors

1. Renal Blood Flow (RBF)

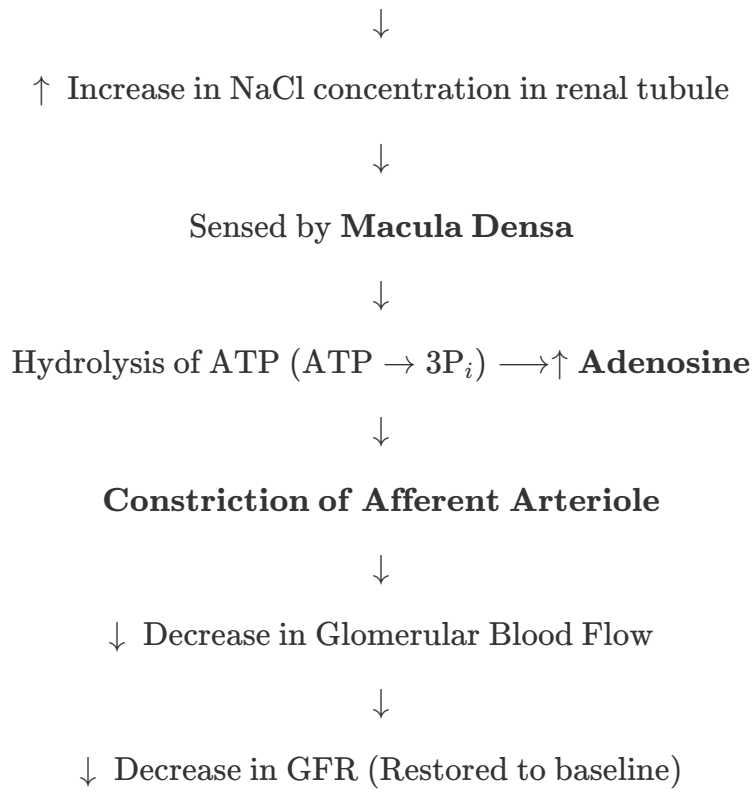
- **Relationship** : $GFR \propto \text{Renal Blood Flow}$.
- **Autoregulation** : Maintained relatively constant across mean arterial pressures of 80 – 180 mmHg via intrinsic renal autoregulatory mechanisms.

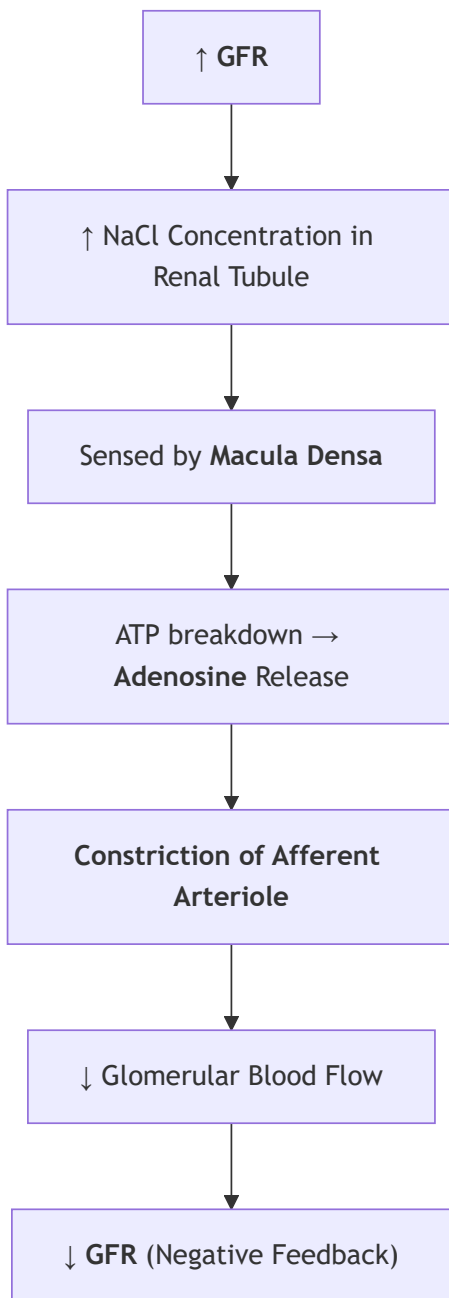
2. Tubuloglomerular Feedback (TGF)

- **Definition** : An intrinsic feedback mechanism that regulates GFR through the renal tubules, specifically mediated by the **macula densa**.
- **Mechanism** : The macula densa senses changes in **sodium chloride (NaCl) concentration** in the tubular filtrate of the distal tubule and adjusts afferent arteriolar resistance accordingly.

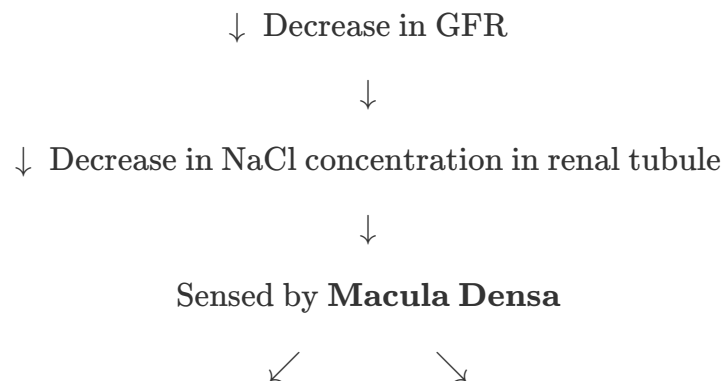
1. When GFR / NaCl conc. inc. in filtrate

↑ Increase in GFR





2. When GFR / NaCl conc. dec. in filtrate



Release of **PGE₂** & **Bradykinin**



Dilation of Afferent Arteriole

Release of **Renin**



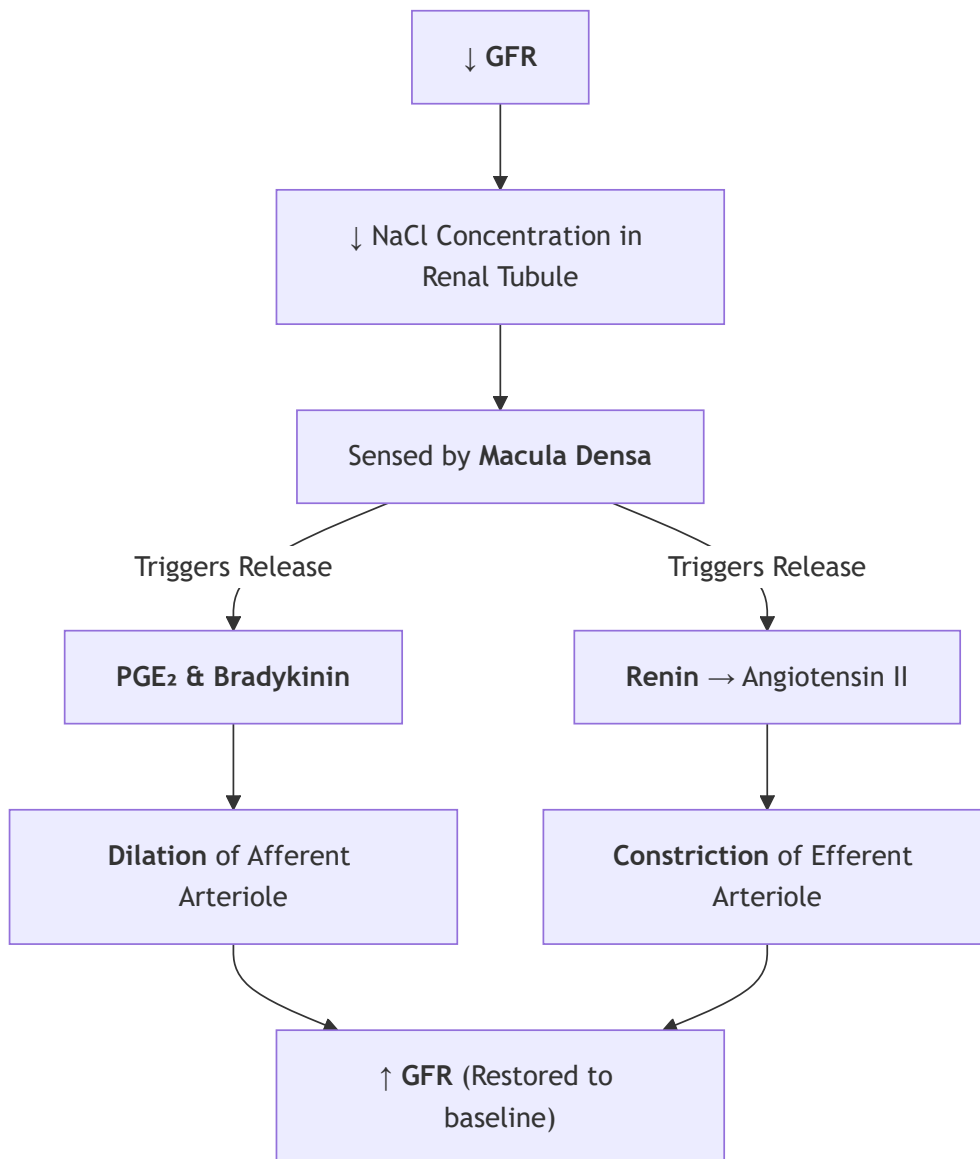
Angiotensinogen → **Angiotensin II**



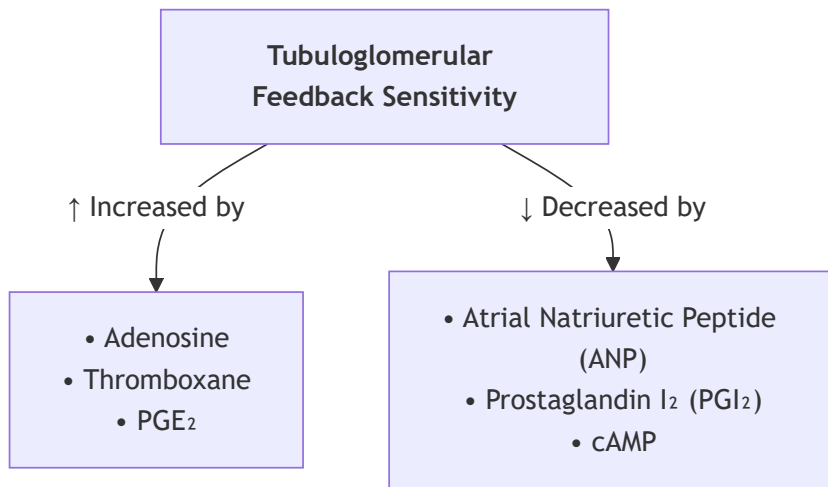
Constriction of Efferent Arteriole



↑ Increase in GFR (Restored to baseline)



Tubuloglomerular Feedback Sensitivity



3. Glomerular Capillary Hydrostatic Pressure (P_{GC})

- **Relationship** : $GFR \propto \text{Glomerular Capillary Pressure}$.
- **Normal Value** : 60 mmHg.
- **Determinants** : Depends directly on **Renal Blood Flow** and **Systemic Arterial Blood Pressure**.

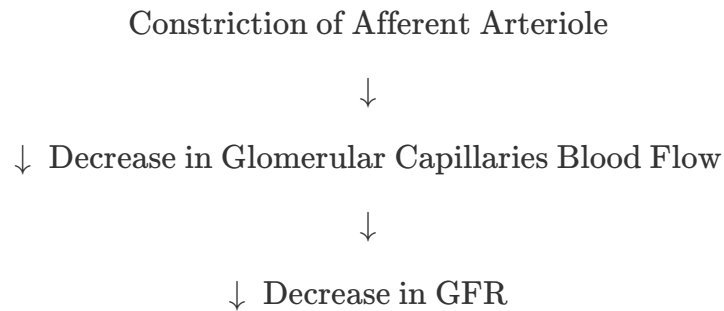
4. Colloid Osmotic / Oncotic Pressure (π_{GC})

- **Relationship** : $GFR \propto \frac{1}{\text{Colloid Osmotic Pressure}}$ (*Inversely proportional*).
- **Normal Value** : 25 mmHg.
- **Variations** :—
 - **↑ Increased in** : Dehydration (concentrates plasma proteins $\longrightarrow$ ↓ decreases GFR).
 - **↓ Decreased in** : Hypoproteinemia (dilutes/lowers plasma protein concentration $\longrightarrow$ ↑ increases GFR).

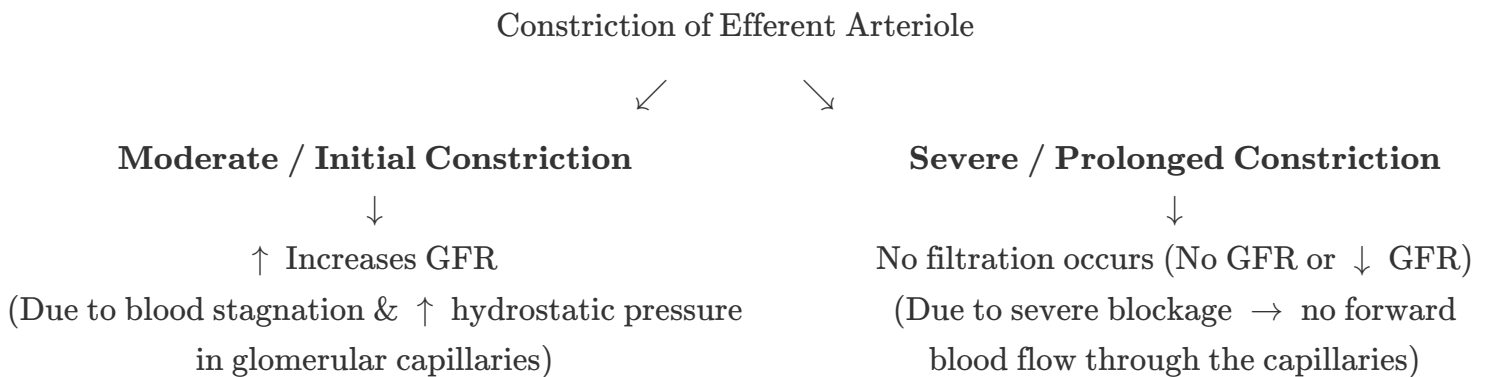
5. Hydrostatic Pressure in Bowman's Capsule (P_{BS})

- **Relationship** : $GFR \propto \frac{1}{\text{Hydrostatic Pressure in Bowman's Capsule}}$ (*Inversely proportional*).
- **↑ Increased in Pathological Conditions** :—
 - Obstruction in the urinary tract (e.g., urethral/ureteral calculi or strictures).
 - Edema of the kidney beneath the non-distensible renal capsule.
 - (*Elevated back-pressure directly opposes filtration $\longrightarrow$ ↓ decreases GFR*).

6. Constriction of Afferent Arteriole



7. Constriction of Efferent Arteriole



8. Systemic Arterial Blood Pressure

- **Autoregulation** : Renal Blood Flow (RBF) and GFR remain relatively **unaffected** as long as the Mean Arterial Blood Pressure stays within the normal autoregulatory range:

60 – 180 mmHg

- **Mechanism** : Regulated intrinsically by renal vascular autoregulation (myogenic mechanism and tubuloglomerular feedback).

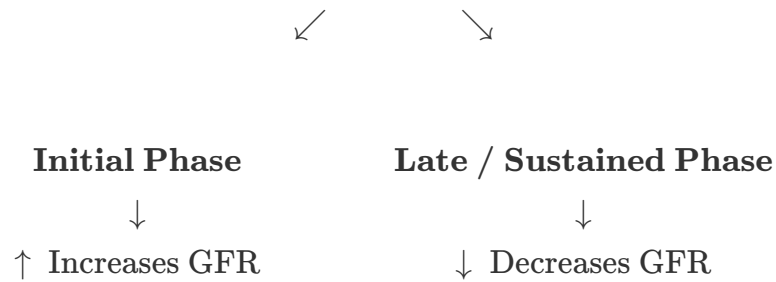
9. Sympathetic Stimulation

- **Moderate Stimulation** : Produces **no significant change in GFR**.
- **Strong Stimulation** :

Strong Sympathetic Stimulation



Severe constriction of Efferent arteriole > Afferent arteriole



10. Surface Area of Capillary Membrane

- Relationship :

$$\text{GFR} \propto \text{Surface Area of Glomerular Capillary Membrane}$$

- ↑ Larger surface area increases GFR; ↓ reduced filtration surface area decreases GFR.

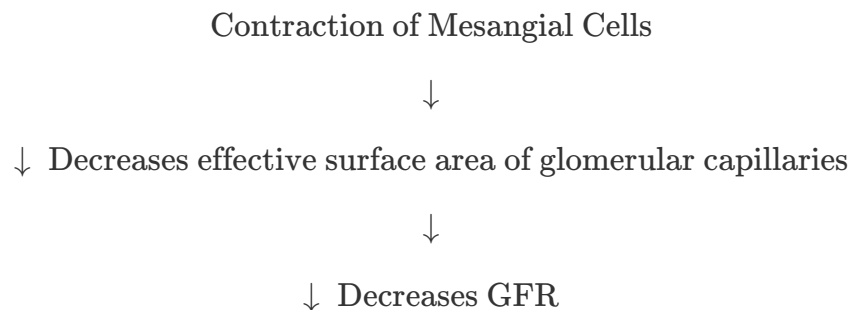
11. Permeability of Capillary Membrane

- Relationship :

$$\text{GFR} \propto \text{Permeability of Capillary Membrane}$$

- ↑ Greater membrane pore permeability increases filtration rate.

12. Contraction of Glomerular Mesangial Cells



13. Hormonal and Other Vasoactive Factors

Factors that ↑ Increase GFR (via Vasodilation)	Factors that ↓ Decrease GFR (via Vasoconstriction)
• Atrial Natriuretic Peptide (ANP) • Brain Natriuretic Peptide (BNP) • Cyclic AMP (cAMP) • Dopamine • Prostaglandin E₂ (PGE₂)	• Angiotensin II • Endothelins • Noradrenaline (Norepinephrine) • Platelet-Activating Factor (PAF) • Prostaglandin F₂ (PGF₂)